

Gradient-Boosted Trees as a Non-Neural Baseline

Rationale, design decisions, and meeting framing

Supervision brief — dissertation extension to Sadhwani, Giesecke & Sirignano (2021)

Prepared for the Friday supervision; preliminary results, full sweep in progress

The dissertation’s central claim — inherited from the paper — is that the monthly transition distribution depends on covariates *nonlinearly and with interactions*, so linear models misspecify. The loan-level result establishes this through the gap between the multinomial logit and the deep network. But that gap conflates two things: *nonlinearity in general* versus *neural networks specifically*. A gradient-boosted-tree (GBT) learner — the standard strong non-neural model for tabular data — separates them. It captures nonlinearity and interactions like the net, but through a different inductive bias (axis-aligned partitions vs. smooth compositions). The paper itself includes no tree baseline, so this comparison on mortgage loan-state transitions is a genuine, low-cost gap to fill, and it sharpens the attribution of the headline result rather than disturbing it.

1 The question GBT answers

The model hierarchy is a deliberately nested ladder of flexibility: empirical matrix $\rightarrow$ bucketed matrix $\rightarrow$ multinomial logit $\rightarrow$ deep network $\rightarrow$ ensemble. The headline finding is that the network beats the logit out of sample in every window, and that nonlinearity supplies roughly two-thirds of the achievable improvement below the covariate-free floor.

The interpretive question a careful reader — or examiner — will press is: *is that gap the value of nonlinearity, or the value of neural networks?* These are not the same claim. The logit-vs-net comparison cannot distinguish them, because only one flexible learner is in the race. Adding a second flexible learner with a *different* inductive bias is what makes the attribution clean.

- **If GBT roughly ties the net:** the gap is about flexibility, not architecture — any sufficiently flexible learner recovers it. This reinforces the existing depth finding (three hidden layers suffice here, not the paper’s five).
- **If the net beats GBT:** the gap reflects something specific to smooth function approximation — a sharper and more interesting claim than “nonlinearity helps.”

Either outcome is a result. That is the point: the comparison is informative *ex ante*, which is what distinguishes it from a fishing expedition.

2 Why gradient-boosted trees specifically

2.1 It is the standard non-neural baseline, not a frontier method

On tabular data of this shape — mixed continuous and categorical features, severe class imbalance, hundreds of millions of rows — gradient boosting is the default-to-beat in the machine-learning literature (the *Elements of Statistical Learning* lineage; the recent tabular-benchmark work of Shwartz-Ziv & Armon, Grinsztajn et al., Borisov et al., and McElfresh et al.). Its *absence* is the first thing an ML-literate reader notices. Framing it as an established statistical-learning baseline — not a novel architecture — keeps the scope honest and the contribution defensible.

2.2 It fills a gap the paper leaves open

The paper benchmarks the deep network against logistic and linear specifications only; it reports no tree or boosting competitor. So the deep-net-vs-GBT comparison *on mortgage loan-state transitions* — and on a clean, public, prime-conforming dataset — has not been done. This is a modest but real “beyond replication” element, and it costs little: one learner through the existing export, loss, evaluation, roll-forward, and cashflow machinery.

3 Why the comparison is genuinely informative: the smoothness hypothesis

This is the intellectual core, and the part worth being able to articulate cleanly.

Gradient-boosted trees approximate a target with axis-aligned, piecewise-constant partitions — a staircase. Neural networks approximate it with smooth compositions. The tabular-ML literature shows that trees hold their largest advantage on *irregular* targets and lose it as the target becomes *smooth* (smoothing a target degrades tree accuracy sharply while barely affecting nets).

The economically important structure in this problem is *smooth*: the prepayment incentive S-curve, the seasoning hump in loan age, the convex decay of delinquency in credit score. That is precisely the regime where the generic tree advantage is weakest and a smooth learner is favoured. So:

Tie $\Rightarrow$ flexibility is what matters, architecture is secondary.

Net edges GBT $\Rightarrow$ the smooth structure of mortgage behaviour rewards a smooth learner — a claim that goes beyond “nonlinearity helps.”

GBT beats net $\Rightarrow$ legitimate, but the dataset metafeatures (enormous N , high row-to-feature ratio, severe imbalance) all favour trees *a priori*, so this would be reported with that caveat rather than overclaimed.

Pre-registering the reading in all three directions is what makes this a test rather than a search for a flattering number.

4 Design decisions and their justification

The single load-bearing decision — the one to be ready to defend — is how GBT is fitted relative to the network.

4.1 One single 7-class model with origin state as a feature — not four per-origin models

The network is a single multinomial model conditioned on the current state as an input feature. GBT matches that exactly. The reason is *experimental control*, and it is the same logic the paper itself uses when it implements the logit as a zero-hidden-layer network so that “the difference is attributable to architecture alone.”

- If GBT instead used four separate per-origin models, a GBT-vs-net difference would confound two things at once: the learner family (trees vs. nets) *and* the model decomposition (four independent fits vs. one joint fit). The clean attribution would be lost.
- The tree-specific reason this costs nothing: origin state has cardinality four, and a single GBT can split on it at any node. So the joint model *contains* the four-separate-models solution as a reachable special case — it can separate where the data justify it and pool where they do not — while four fixed models impose full separation everywhere. Under the severe origin imbalance (the delinquent origins are orders of magnitude sparser than **current**), the joint model’s ability

to pool statistical strength is an advantage, not a compromise.

4.2 The remaining choices, each in one line

- **Identical feature set, minus standardisation.** GBT is scale-invariant, so the per-window scaler is skipped; every other feature column is the same as the net’s, so the comparison isolates the learner.
- **Native categorical handling** on the same vocabulary codes the embeddings use (unseen levels routed to a reserved index) — not one-hot, not embeddings.
- **Importance weights on the training set only.** The Horvitz–Thompson inverse-probability weights enter as per-row sample weights, so the thinned-sample loss stays an unbiased estimate of the full-sample loss, exactly as in the network’s objective. Validation and test are unthinned and unweighted.
- **Multiclass softmax objective** (= multinomial log-loss = the headline NLL), so early stopping optimises the reported quantity.
- **Tune once on the tuning window, freeze, roll** — the same protocol as the nets, avoiding per-window re-tuning and its multiple-testing problem.
- **CPU, not GPU** — a deliberate, documented deviation: at this feature count GBT is CPU-bound and the GPU histogram speedup is negligible, so it runs where the other non-neural baselines run.
- **Calibration is conditional** — the one step the nets did not need. Because the pool-level valuation consumes probability *levels*, if the reliability diagrams show GBT is miscalibrated, a per-window post-hoc calibration is fitted on the validation slice before the roll-forward.

5 A coherent secondary thread: prime credit is structurally simpler

GBT slots into a narrative the dissertation already tells. The architecture search selected *three* hidden layers rather than the paper’s five — the nonlinearity premium is real but structurally simpler in the prime conforming credit box. The GBT tuning echoes this from the tree side: at full scale the optimum is a *shallow, heavily leaf-regularised* learner (31 leaves), and the leaf-regularisation strength had to scale up roughly tenfold from the development slice precisely because the importance weights inflate the summed leaf curvature at scale — the tree-side analogue of “three layers, not five.” Both learners, independently, report that this credit box has a smoother, simpler target surface than the subprime-heavy data of the original study.

6 Preliminary evidence (loan level, two of eleven windows)

Two regime windows have completed. Both are healthy and both point the same way: GBT clearly beats the logit, and the network edges GBT by a small margin — the “flexibility wins, with a mild lean to the smooth learner” reading.

Window	GBT NLL	vs. logit	vs. net	rounds	QA
k=2015 (calm)	0.103597	−0.00465	+0.00047	1208	all pass
k=2019 (late cycle)	0.103496	−0.00286	+0.00290	130	impossible-mass flag

Reference points at full scale (k=2015): empirical floor 0.1129, bucketed matrix 0.1094, logit 0.1082, network 0.1031. GBT lands between the logit and the network, close to the net — and the gap to the net *compressed* relative to the development-scale fit, a detail worth a line in the write-up.

- **On the lone QA flag.** $k=2019$'s predicted mass on structurally-impossible cells (1.86×10^{-3}) sits just above the 10^{-3} threshold. This is almost certainly a fit-maturity artifact: that window early-stopped at only 130 rounds, so the model has not yet driven near-impossible cells to zero. The prediction to confirm across the full sweep is that only the lightly-fit (low-round) windows breach and the long-fit ones stay clean — in which case it is a calibration footnote, recorded, not tuned away. A breach on a *long*-fit window would be a different signal worth investigating.
- **Honest status.** Two windows are calm/late-cycle. The window that will actually test the smoothness lean is the COVID-onset window, where the inductive biases are most likely to diverge — and it runs first in the remaining sweep.

7 Framing for the meeting

“I’ve added a gradient-boosted-tree baseline alongside the network — the standard non-neural learner for this kind of tabular data — to separate the value of *nonlinearity* from the value of *neural networks specifically*. It goes through exactly the same machinery as the net, so any gap is attributable to inductive bias alone. Early results have it clearly beating the logit and sitting just behind the net, which is the ‘flexibility wins, smooth target mildly favours the net’ reading — but it’s two of eleven windows so far.”

7.1 What to lead with

1. **The control logic first.** It is the paper’s own logic (the logit-as-zero-layer-net device), so it signals you have internalised the methodology rather than bolted something on. “Same export, same loss, same evaluation — the only thing that varies is the learner” is the sentence that establishes rigour.
2. **The gap it fills, framed as strengthening, not correcting.** The paper benchmarks against linear models; adding the standard non-neural flexible learner sharpens the attribution of the existing result. It is additive to the paper’s story, not a critique of it.
3. **The smoothness hypothesis as the genuinely interesting part.** This is the idea most worth his time: the prime-conforming setting, with its smooth incentive S-curve and seasoning hump, is exactly where a smooth learner should be favoured over a staircase-fitting one — so the comparison is a real test of *why* flexibility helps, not just *that* it does.
4. **The “simpler in prime credit” thread.** Tie it to the three-layer finding he already knows. Both learners independently report a simpler target surface than the subprime data — a coherent secondary narrative, not an isolated add-on.

7.2 Anticipated questions, and crisp answers

- **“Why GBT and not another architecture, or a GAM?”** GBT is the standard *non-neural* flexible learner; the point is a *different inductive bias*, not more flexibility of the same kind. A GAM is largely redundant here because the model-free hazard curves already isolate the interactions a GAM would test.
- **“Does this change the headline result?”** No — it strengthens its interpretation. The network still beats the linear models; GBT tells us whether that edge is nonlinearity-in-general or something specific to smooth approximation.
- **“Is it calibrated enough for the pool-level valuation?”** The pool translation consumes levels, so calibration is checked explicitly via reliability diagrams; if GBT is off-diagonal, a per-

window post-hoc calibration is applied before the roll-forward. The nets did not need this; GBT may.

- **“Isn’t this scope creep for a dissertation?”** It is one learner through the existing pipeline — cheap relative to its interpretive payoff, and it produces a clean result whichever way it lands.
- **“What if GBT beats the net?”** Pre-registered as a legitimate outcome, but read with the caveat that the dataset’s size and imbalance favour trees *a priori*; disentangling that from a true edge would call for the smoothness ablation.

7.3 What to be candid about

- Only two of eleven windows are in, both from calm/late-cycle regimes; the regime-defining windows (COVID, the rate shock) are pending.
- The pool-level translation for GBT — the economically meaningful comparison — is not yet run; the current evidence is loan-level likelihood only.
- The single QA flag is logged as a finding to verify, not smoothed over.

“Before I price GBT through to the pool level, does the framing hold up — in particular, is the smoothness hypothesis the right way to read a tie-or-near-tie, and is there a regime you’d most want me to look at first?” This invites him into the interpretation, which is where a co-author of the original paper adds the most value — and turns the meeting from a status report into a design conversation.

Status at time of writing: two loan-level windows banked, mirrored, and committed; the remaining nine windows (COVID-onset first) queued for a high-RAM run; pool-level translation to follow. All numbers above are out-of-sample mean negative log-likelihood on frozen test rows; lower is better.